

Dynamic Scene Graphs for Action Recognition in Egocentric and Exocentric Videos

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Abstract

Action recognition in dynamic environments remains a challenging problem due to occlusions, scene complexity, and multi-object interactions. The use of scene graphs has introduced a new paradigm when perceiving the environment in comparison to static dense representations. It allows for efficient parsing of the environment into a dynamically extendable structure. We would like to extend scene graphs with multi-object interactions that happen often in human life. We plan on representing three-dimensional video data in the form of scene graphs. Once constructed, we will perform semantic segmentation to identify objects within our scene graph. We aim to update these scene graphs dynamically with temporal graph networks to link objects with actions. Our proposed dual graph model will link objects to their distances within physical space and to their relationships within actions.

1. Scene Graph Generation

The 3D Scene Graph generation task can be divided into two stages.

1. Processing the input data and obtaining the semantic instance classified representation of the scene.
2. Creating a graph to represent the processed data with further annotations to be useful for downstream tasks

1.1. Understanding the Sceme

1.1.1 Keyframe Based Scene Understanding

One example of this approach is OpenMask3d[10]. That uses a sparse convolutional point cloud instance segmentation network with CLIP for open-vocabulary image embedding. The structure of this method is also more suited for dynamically generated scenes. Where the incoming image stream can be incrementally processed for new objects and used to update the map.

Main drawback of using this method is added complexity in comparison to fully point cloud based methods.

1.1.2 Point Cloud Based Scene Understanding

One example of this approach is Mask3d[9], which is built upon a sparse convolutional point cloud feature network. It directly works on the point-cloud level and does not require an intermediate semantic segmentation projection step. Biggest limitation of this approach is being limited to closed vocabulary feature networks. Which reduces the generalizability of the method in diverse task configurations. It is also assumed that the full point cloud of the scene is available during data processing, which is hardly the case in dynamical use cases such as robotics.

1.2. Generating a 3D Scene Graph

Instead of representing the objects and the scene with binary segmentation masks or per point features, more efficient and useful representations are required for downstream tasks, i.e. motion-planning. One prominent example is Concept-Graphs [4]. Which generated from RGB-D images a 3DSGs with nodes containing geometric information and captions generated by a VLM and edges representing the spatial information of the scheme as a minimum spanning tree.

1.3. Extending the 3D Scene Graph with Actions

We are proposing the unified action-object structure in 1. It consists of a root World node for spatially representing the scene with relative transformation edges and multiple action root nodes describing the action and action edges connecting the action node with the interaction objects. This action graph structure allows for multiple ongoing actions and spatial grounding of the actions. The structure can also be extended with more informative action edges specifying the type of interaction happening for further use cases.

2. Detecting and Interpreting Actions

2.1. Frame-by-Frame Scene Graph Updates

In each time step or keyframe we can segment objects using OpenMask3d[10] or Mask3d [9] and update the Concept-Graph[2] with newly seen or removed objects. Updating the scene graphs in the egocentric case can be done using [1]. A short temporal window classifier or rule-based inference

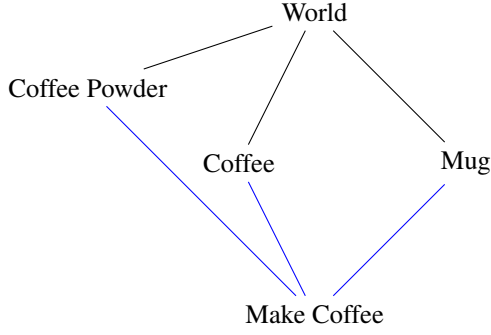


Figure 1. The dual graph structure represents actions and objects together. The black edges represent relative poses in the scene and the blue edges represent semantic relations between an action and the objects used during this action

can decide which action label applies (e.g., grabbing, placing, pouring) based on how close a human node is to certain objects. This approach is similar to existing 2D methods (e.g. Action Genome [5]) where scene graphs in each frame are used to label interactions. The drawbacks are that it is harder for this method to handle long or complex actions with multiple sub-steps.

2.2. Action Recognition and Understanding (With TGNNs)

Instead of frame-by-frame heuristics, Temporal Graph Neural Networks (TGNNs) let us learn node embeddings that evolve over time (e.g., DynGESN [6], EvolveGCN [8], TGN [7]). Each snapshot or event updates the scene graph which allows capturing multi-step or parallel actions (“grabbing,” “pouring,” etc.) more effectively than simpler methods [5]. By integrating our 3D scene graph with a TGNN, we can track how objects and people link together over time, yielding more precise detection of complex or overlapping tasks.

3. Temporal Graph Networks

In order to detect actions in continuous video frames, we see two possible ways.

1. We use temporal graph classification, which seems straightforward but only allows one to classify one single action at any given point in time. A possible model is DynGESN [6].
2. We use temporal link prediction and predict that each object involved in an action sequence will be connected to a predicted action node (see Figure 1). A possible model is EvolveGCN [8].

Furthermore, we will need to decide whether we want to use a snapshot-based or an event-based model. A snapshot-based model processes a graph in a fixed interval and processes the entire graph, while an event-based model only

updates the necessary parts of the graph based on events [7].

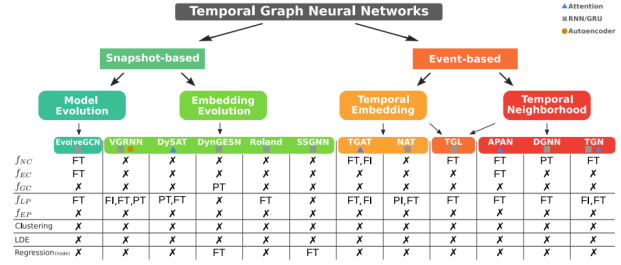


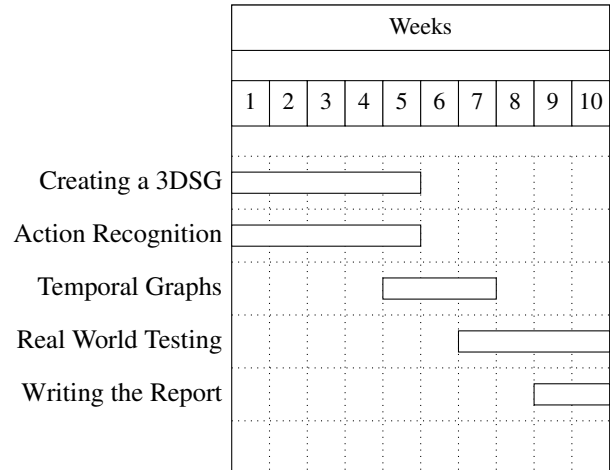
Figure 2. An overview of a temporal graph neural networks classification and corresponding models [7]

4. Dataset

The dataset Ego-Exo-4d [3] includes synchronized images from 3rd perspective and first-perspective data from project-aria glasses. It includes trajectories and poses of the images and has preprocessed point-clouds of the environment to be used for scene generation. It also includes human annotations describing the action happening in the first person view.

One of the main advantages of this dataset is creating a pipeline working with aria glasses allows for future improvements and testing in the real world.

5. Timeline and Tasks



6. Risks

6.1. Dataset

Main disadvantage of the Ego-Exo-4d dataset is the amount of complexity present on some of the scenes. The methods presented for scene understanding has not yet been applied to task settings with highly cluttered environments

and might pose a challenge. To cope with this challenge we might have to generate our own sample datasets in controlled environments for development stage of the project.

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